La Scuola d'Italia / Dr. Huson / IB Math (a)
29 May 2026

First & Last name:

8.5 Quiz: Rational Functions and Transformations (no calculator)

1. [Maximum mark: 8]

   Consider the function $f(x) = \dfrac{2}{x-3} + 1$.

   (a) Write down the equations of the vertical and horizontal asymptotes of the graph of f. [2]

   (b) Describe fully the sequence of transformations that maps the graph of $y = \dfrac{1}{x}$ onto the graph of f. [3]

   (c) Find the y-intercept of the graph of f. [1]

   (d) Find the x-intercept of the graph of f. [2]

2. [Maximum mark: 8]

The function g is defined by $g(x) = \dfrac{2x+1}{x-1}$, $x \in \mathbb{R}$, $x \neq 1$.

(a) Write down the equations of the vertical and horizontal asymptotes of the graph of g. [2]

(b) Find the coordinates of the x-intercept and of the y-intercept of the graph of g. [2]

(c) Find $g^{-1}(x)$. [3]

(d) State the domain of g^{-1}. [1]

La Scuola d'Italia / Dr. Huson / IB Math (b)
29 May 2026

First & Last name:

8.5 Quiz: Rational Functions and Transformations (no calculator)

1. [Maximum mark: 8]

   Consider the function $f(x) = \dfrac{3}{x+2} - 1$.

   (a) Write down the equations of the vertical and horizontal asymptotes of the graph of f. [2]

   (b) Describe fully the sequence of transformations that maps the graph of $y = \dfrac{1}{x}$ onto the graph of f. [3]

   (c) Find the y-intercept of the graph of f. [1]

   (d) Find the x-intercept of the graph of f. [2]

2. [Maximum mark: 8]

The function g is defined by $g(x) = \dfrac{3x-2}{x+1}$, $x \in \mathbb{R}$, $x \neq -1$.

(a) Write down the equations of the vertical and horizontal asymptotes of the graph of g. [2]

(b) Find the coordinates of the x-intercept and of the y-intercept of the graph of g. [2]

(c) Find $g^{-1}(x)$. [3]

(d) State the domain of g^{-1}. [1]

La Scuola d'Italia / Dr. Huson / IB Math (c)
29 May 2026

First & Last name:

8.5 Quiz: Rational Functions and Transformations (no calculator)

1. [Maximum mark: 8]

   Consider the function $f(x) = \dfrac{4}{x-1} + 2$.

   (a) Write down the equations of the vertical and horizontal asymptotes of the graph of f. [2]

   (b) Describe fully the sequence of transformations that maps the graph of $y = \dfrac{1}{x}$ onto the graph of f. [3]

   (c) Find the y-intercept of the graph of f. [1]

   (d) Find the x-intercept of the graph of f. [2]

2. [Maximum mark: 8]

The function g is defined by $g(x) = \dfrac{4x+3}{x-2}$, $x \in \mathbb{R}$, $x \neq 2$.

(a) Write down the equations of the vertical and horizontal asymptotes of the graph of g. [2]

(b) Find the coordinates of the x-intercept and of the y-intercept of the graph of g. [2]

(c) Find $g^{-1}(x)$. [3]

(d) State the domain of g^{-1}. [1]